Step by Step Data Guard Configuration in 19c

Enterprise Oracle Database 19c Physical Standby Setup & Synchronization Architecture

1. PREREQUISITES

Requirements and environment specifications before starting the Data Guard configuration:

Parameter / Details	Primary Database (DG11)	Standby Database (DG11S)
Server Name	DG11	DG11S
IP Address	192.168.56.10	192.168.56.11
Hostname	dg11.localdomain	dg11s.localdomain
Global DB Name	DG11	DG11
DB_UNIQUE_NAME	DG11	DG11S
SID Name	DG11	DG11S

Prerequisite Check Verification:

- **Network Connectivity:** Ping tests completed successfully between both nodes:
  `ping 192.168.56.10` (Primary server)
  `ping 192.168.56.11` (Standby server)
- **Software Installation:** Oracle Database 19c Grid / Software binaries installed on both servers.
- **Primary Database:** Created and running in Read/Write mode.
- **Standby Environment:** Standby server pre-configured with hostname, IP, and directories.

2. COMPLETE STEP BY STEP CONFIGURATION FOR PRIMARY DATABASE

High-level steps required to configure the Primary DB:

1. Create LISTENER for PRIMARY DB and start the LISTENER by DISABLING FIREWALL.
2. Create TNSNAMES.ORA file for PRIMARY DB and check TNSPING for PRIMARY DB and STANDBY DB.
3. ENABLE ARCHIVE LOG for Primary DB.
4. ENABLE FORCE LOGGING for Primary DB.
5. Check DB_NAME and DB_UNIQUE_NAME for Primary DB.
6. Set LOG_ARCHIVE_CONFIG parameter for Primary DB.
7. Set LOG_ARCHIVE_DEST parameter for Primary DB.
8. Set LOG_ARCHIVE_DEST_STATE parameter to ENABLE for Primary DB.

9. Set LOG_ARCHIVE_FORMAT parameter for Primary DB.
10. Set LOG_ARCHIVE_MAX_PROCESSES parameter to 30 for Primary DB.
11. Set REMOTE_LOGIN_PASSWORDFILE parameter to EXCLUSIVE for Primary DB.
12. Set FAL_SERVER and FAL_CLIENT parameters for Primary DB.
13. Set STANDBY_FILE_MANAGEMENT parameter to MANUAL for Primary DB.
14. Add STANDBY REDOLOG FILES (SRLs) to Primary DB while STANDBY_FILE_MANAGEMENT=MANUAL.
15. Set STANDBY_FILE_MANAGEMENT parameter back to AUTO for Primary DB.
16. Send PASSWORD FILE from PRIMARY DB to STANDBY DB.

STEP 1: Create LISTENER and Disable Firewall

A) Configure listener.ora on Primary DB: Path: `/u01/app/oracle/product/19.0.0/dbhome_1/network/admin/listener.ora`

```
SID_LIST_LISTENER =
  (SID_LIST =
    (SID_DESC =
      (GLOBAL_DBNAME = dg11)
      (ORACLE_HOME = /u01/app/oracle/product/19.0.0/dbhome_1)
      (SID_NAME = dg11)
    )
  )

LISTENER =
  (DESCRIPTION =
    (ADDRESS = (PROTOCOL = TCP)(HOST = 192.168.56.10)(PORT = 1521))
  )

ADR_BASE_LISTENER = /u01/app/oracle
```

B) Disable Firewall on Primary & Standby Servers:

```
[oracle@localhost admin]$ su root
Password:
[root@localhost admin]# service firewalld stop
Redirecting to /bin/systemctl stop firewalld.service
[root@localhost admin]# systemctl disable firewalld
[root@localhost admin]# service firewalld status
● firewalld.service - firewalld - dynamic firewall daemon
   Loaded: loaded (/usr/lib/systemd/system/firewalld.service; disabled; vendor preset:
enabled)
   Active: inactive (dead)
```

C & D) Start Listener on Primary & Standby DBs:

```
[oracle@localhost admin]$ lsnrctl start
```

STEP 2: Create TNSNAMES.ORA and Check TNSPING

Create tnsnames.ora on Primary DB: Path: `/u01/app/oracle/product/19.0.0/dbhome_1/network/admin/tnsnames.ora`

```
DG11 =
  (DESCRIPTION =
    (ADDRESS_LIST =
      (ADDRESS = (PROTOCOL = TCP)(HOST = 192.168.56.10)(PORT = 1521))
    )
    (CONNECT_DATA =
      (SERVICE_NAME = dg11)
    )
  )

DG11S =
  (DESCRIPTION =
    (ADDRESS_LIST =
      (ADDRESS = (PROTOCOL = TCP)(HOST = 192.168.56.11)(PORT = 1521))
    )
    (CONNECT_DATA =
      (SERVICE_NAME = dg11s)
    )
  )
```

Verify Connectivity with TNSPING:

```
[oracle@localhost admin]$ tnsping dg11
Attempting to contact (DESCRIPTION = (ADDRESS_LIST = (ADDRESS = (PROTOCOL = TCP)(HOST =
192.168.56.10)(PORT = 1521))) (CONNECT_DATA = (SERVICE_NAME = dg11)))
OK (0 msec)

[oracle@localhost admin]$ tnsping dg11s
Attempting to contact (DESCRIPTION = (ADDRESS_LIST = (ADDRESS = (PROTOCOL = TCP)(HOST =
192.168.56.11)(PORT = 1521))) (CONNECT_DATA = (SERVICE_NAME = dg11s)))
OK (10 msec)
```

STEP 3: Enable ARCHIVE LOG for Primary DB

Check archive log status:

```
SQL> select name, open_mode, database_role from v$database;
NAME      OPEN_MODE            DATABASE_ROLE
--------  -------------------- ----------------
DG11      READ WRITE           PRIMARY

SQL> select log_mode from v$database;
LOG_MODE
------------
ARCHIVELOG
```

> *Note:* If `LOG_MODE` is `NOARCHIVELOG`, execute:
> `SQL> SHUTDOWN IMMEDIATE;`
> `SQL> STARTUP MOUNT;`
> `SQL> ALTER DATABASE ARCHIVELOG;`
> `SQL> ALTER DATABASE OPEN;`

STEP 4: Enable FORCE LOGGING for Primary DB

```
SQL> select FORCE_LOGGING from v$database;
FORCE_LOGGING
---------------------------------
NO

SQL> alter database FORCE LOGGING;
Database altered.

SQL> select FORCE_LOGGING from v$database;
FORCE_LOGGING
---------------------------------
YES
```

STEP 5: Check DB_NAME and DB_UNIQUE_NAME

```
SQL> show parameter DB_NAME;
NAME                                 TYPE        VALUE
------------------------------------ ----------- ------------------------------
db_name                              string      DG11

SQL> show parameter DB_UNIQUE_NAME;
NAME                                 TYPE        VALUE
------------------------------------ ----------- ------------------------------
db_unique_name                       string      DG11
```

STEP 6 to 13: Configure Initialization Parameters on Primary DB

STEP 6: SET LOG_ARCHIVE_CONFIG

```
SQL> alter system set LOG_ARCHIVE_CONFIG='DG_CONFIG=(dg11,dg11s)';
System altered.
```

STEP 7: SET LOG_ARCHIVE_DEST_2

```
SQL> alter system set LOG_ARCHIVE_DEST_2='SERVICE=dg11s NOAFFIRM ASYNC
VALID_FOR=(ONLINE_LOGFILES,PRIMARY_ROLE) DB_UNIQUE_NAME=dg11s';
System altered.
```

STEP 8: SET LOG_ARCHIVE_DEST_STATE_2

```
SQL> alter system set LOG_ARCHIVE_DEST_STATE_2=enable;
System altered.
```

STEP 9: SET LOG_ARCHIVE_FORMAT

```
SQL> alter system set LOG_ARCHIVE_FORMAT='%t_%s_%r.arc' scope=spfile;
System altered.
```

STEP 10: SET LOG_ARCHIVE_MAX_PROCESSES

```
SQL> alter system set LOG_ARCHIVE_MAX_PROCESSES=30;
System altered.
```

STEP 11: SET REMOTE_LOGIN_PASSWORDFILE

```
SQL> alter system set REMOTE_LOGIN_PASSWORDFILE=exclusive scope=spfile;
System altered.
```

STEP 12: SET FAL_CLIENT AND FAL_SERVER

```
SQL> alter system set FAL_CLIENT=DG11;
System altered.
SQL> alter system set FAL_SERVER=DG11S;
System altered.
```

STEP 13: SET STANDBY_FILE_MANAGEMENT TO MANUAL (BEFORE ADDING SRLS)

```
SQL> alter system set STANDBY_FILE_MANAGEMENT=MANUAL;
System altered.
```

STEP 14: Add Standby Redo Log Files (SRLs)

First, check existing Online Redo Logs to ensure group numbers do not overlap and sizes match (200MB):

```
SQL> select a.GROUP#, a.THREAD#, a.BYTES/1024/1024, a.MEMBERS, a.SEQUENCE#, a.STATUS,
b.TYPE, b.MEMBER
     from v$log a, v$logfile b where a.GROUP# = b.GROUP#;

    GROUP#    THREAD#  A.BYTES/1024/1024    MEMBERS  SEQUENCE# STATUS   TYPE    MEMBER
---------- ---------- ------------------ ---------- ---------- -------- -------
--------------------------------------
         3          1                200          1          6 INACTIVE ONLINE /u01/app/
oracle/oradata/DG11/redo03.log
         2          1                200          1          5 INACTIVE ONLINE /u01/app/
oracle/oradata/DG11/redo02.log
         1          1                200          1          7 CURRENT  ONLINE /u01/app/
oracle/oradata/DG11/redo01.log
```

Add 3 Standby Redo Log groups (Group 4, 5, 6) of size 200M:

```
SQL> ALTER DATABASE ADD STANDBY LOGFILE THREAD 1 GROUP 4 ('/u01/app/oracle/oradata/DG11/
standby_redo04.log') SIZE 200M;
SQL> ALTER DATABASE ADD STANDBY LOGFILE THREAD 1 GROUP 5 ('/u01/app/oracle/oradata/DG11/
standby_redo05.log') SIZE 200M;
SQL> ALTER DATABASE ADD STANDBY LOGFILE THREAD 1 GROUP 6 ('/u01/app/oracle/oradata/DG11/
standby_redo06.log') SIZE 200M;
```

Verify Standby Redo Logs creation:

```
SQL> SELECT GROUP#, THREAD#, SEQUENCE#, ARCHIVED, STATUS FROM V$STANDBY_LOG;

    GROUP#    THREAD#  SEQUENCE# ARC STATUS
---------- ---------- ---------- --- ----------
         4          1          0 YES UNASSIGNED
         5          1          0 YES UNASSIGNED
         6          1          0 YES UNASSIGNED
```

STEP 15: Set STANDBY_FILE_MANAGEMENT to AUTO

```
SQL> alter system set STANDBY_FILE_MANAGEMENT=AUTO;
System altered.
```

STEP 16: Send Password File to Standby DB

```
[oracle@localhost dbs]$ cd /u01/app/oracle/product/19.0.0/dbhome_1/dbs/
[oracle@localhost dbs]$ scp orapwDG11 oracle@192.168.56.11:/u01/app/oracle/product/
19.0.0/dbhome_1/dbs/orapwDG11s
oracle@192.168.56.11's password:
orapwDG11                                              100% 2048     1.3MB/s
```

3. COMPLETE STEP BY STEP CONFIGURATION FOR STANDBY DATABASE

High-level steps required to configure the Standby DB:

1. Make entry in ORATAB file before restoring Standby DB.
2. Create LISTENER for STANDBY DB and start the LISTENER by DISABLING FIREWALL.
3. Create TNSNAMES.ORA file for STANDBY DB and check TNSPING for PRIMARY DB and STANDBY DB.
4. Set the ENVIRONMENT of Standby DB for 1st time by ECHOING/oraenv.
5. Create a P-FILE for Standby DB and create required DIRECTORIES.
6. STARTUP the DB in NOMOUNT mode using new P-FILE created for Standby DB.
7. Connect to AUXILIARY DB using RMAN PROMPT.
8. RUN the SCRIPT in RMAN PROMPT to CLONE Primary DB to Standby DB.
9. STARTUP the DB in READ_ONLY mode and validate/rectify parameters.

STEP 1: Make Entry in /etc/oratab

```
[oracle@localhost dbs]$ cat /etc/oratab
DG11S:/u01/app/oracle/product/19.0.0/dbhome_1:N
```

STEP 2 & 3: Configure Listener and TNSNAMES on Standby Server

Listener Configuration: Path: `/u01/app/oracle/product/19.0.0/dbhome_1/network/admin/listener.ora`

```
SID_LIST_LISTENER =
  (SID_LIST =
    (SID_DESC =
      (GLOBAL_DBNAME = DG11S)
      (ORACLE_HOME = /u01/app/oracle/product/19.0.0/dbhome_1)
      (SID_NAME = DG11S)
    )
  )

LISTENER =
  (DESCRIPTION =
    (ADDRESS = (PROTOCOL = TCP)(HOST = 192.168.56.11)(PORT = 1521))
  )

ADR_BASE_LISTENER = /u01/app/oracle
```

Start listener via `lsnrctl start` and verify firewall is disabled.

STEP 4: Set Environment on Standby Host

```
[oracle@localhost admin]$ . oraenv
ORACLE_SID = [oracle] ? DG11S
The Oracle base has been set to /u01/app/oracle
```

STEP 5: Create Initialization P-FILE & Directories for Standby DB

A) Create minimal initDG11S.ora: Path: `/u01/app/oracle/product/19.0.0/dbhome_1/dbs/initDG11S.ora`

```
*.db_name='DG11'
```

B) Create required admin and data directories on Standby:

```
[oracle@localhost u01]$ mkdir -p /u01/app/oracle/admin/dg11/adump
[oracle@localhost u01]$ mkdir -p /u01/app/oracle/oradata/DG11
[oracle@localhost u01]$ mkdir -p /u01/app/oracle/fast_recovery_area/DG11/
```

> *Important Note:* Directories use the GLOBAL DB NAME 'DG11' rather than DB_UNIQUE_NAME 'DG11S'. This ensures RMAN DUPLICATE path translation succeeds without missing directory errors.

STEP 6: Start Standby Instance in NOMOUNT Mode

```
SQL> startup nomount;
ORACLE instance started.

Total System Global Area   243268216 bytes
Fixed Size                   8895096 bytes
Variable Size              180355072 bytes
Database Buffers            50331648 bytes
Redo Buffers                 3686400 bytes
```

STEP 7: Connect to Auxiliary DB via RMAN Prompt

```
[oracle@localhost ~]$ rman target sys/oracle@DG11 auxiliary sys/oracle@DG11S

Recovery Manager: Release 19.0.0.0.0 - Production on Fri Aug 21 15:54:54 2026
Version 19.3.0.0.0

connected to target database: DG11 (DBID=954793642)
connected to auxiliary database: DG11 (not mounted)
```

STEP 8: Duplicate Target Database for Standby

Execute the following RMAN script to clone Primary DB to Standby DB:

```
run
{
  DUPLICATE TARGET DATABASE FOR STANDBY FROM ACTIVE DATABASE
  SPFILE
    PARAMETER_VALUE_CONVERT='DG11','DG11S'
    SET DB_NAME='DG11'
    SET DB_UNIQUE_NAME='DG11S'
    SET STANDBY_FILE_MANAGEMENT='AUTO'
    SET audit_file_dest='/u01/app/oracle/admin/dg11/adump'
    SET db_recovery_file_dest='/u01/app/oracle/fast_recovery_area/'
    SET fal_server='DG11'
    SET fal_client='DG11S'
    SET control_files='/u01/app/oracle/oradata/DG11/control01.ctl','/u01/app/oracle/
fast_recovery_area/DG11/control02.ctl'
    NOFILENAMECHECK;
}
```

STEP 9: Open Standby in READ ONLY Mode & Rectify Parameters

```
SQL> alter database open;
Database altered.

SQL> select name, open_mode, DB_UNIQUE_NAME, DATABASE_ROLE from v$database;
NAME      OPEN_MODE            DB_UNIQUE_NAME                 DATABASE_ROLE
--------  -------------------- ------------------------------ ----------------
DG11      READ ONLY            DG11S                          PHYSICAL STANDBY
```

Rectify log_archive_dest_2 on Standby DB:

Reset parameter:

```
SQL> ALTER SYSTEM SET LOG_ARCHIVE_DEST_2='' scope=both;
System altered.
```

Set correct destination pointing back to Primary (DG11):

```
SQL> alter system set LOG_ARCHIVE_DEST_2='SERVICE=DG11 NOAFFIRM ASYNC
VALID_FOR=(ONLINE_LOGFILES,PRIMARY_ROLE) DB_UNIQUE_NAME=DG11';
System altered.
```

4. COMPLETE PROCESS FOR MAKING PRIMARY & STANDBY IN SYNC

High-level steps for validating and starting log apply synchronization:

1. Check the MAXIMUM SEQUENCE of ARCHIVED LOGS generated/applied in Primary and Standby DB.
2. START the MRP (Managed Recovery Process) in STANDBY DB for applying archives.
3. Check whether the remaining archived logs are applied by MRP to Standby DB.

STEP 1: Check Archive Sequences

On PRIMARY DB (DG11): Switch logfiles to generate new archive logs.

```
SQL> select MAX(SEQUENCE#) from v$archived_log;
MAX(SEQUENCE#)
--------------
             7

SQL> alter system switch logfile;
System altered.

SQL> alter system switch logfile;
System altered.

SQL> select MAX(SEQUENCE#) from v$archived_log;
MAX(SEQUENCE#)
--------------
             9
```

On STANDBY DB (DG11S): Verify received logs from Primary.

```
SQL> select MAX(SEQUENCE#) from v$archived_log;
MAX(SEQUENCE#)
--------------
             9
```

Check log gap status on Standby DB using query:

```
SQL> SELECT ARCH.THREAD# "Thread",
       ARCH.SEQUENCE# "Last Sequence Received",
       APPL.SEQUENCE# "Last Sequence Applied",
       (ARCH.SEQUENCE# - APPL.SEQUENCE#) "Difference"
FROM
  (SELECT THREAD#, SEQUENCE# FROM V$ARCHIVED_LOG WHERE (THREAD#, FIRST_TIME) IN
    (SELECT THREAD#, MAX(FIRST_TIME) FROM V$ARCHIVED_LOG GROUP BY THREAD#)) ARCH,
  (SELECT THREAD#, SEQUENCE# FROM V$LOG_HISTORY WHERE (THREAD#, FIRST_TIME) IN
    (SELECT THREAD#, MAX(FIRST_TIME) FROM V$LOG_HISTORY GROUP BY THREAD#)) APPL
WHERE ARCH.THREAD# = APPL.THREAD# ORDER BY 1;

    Thread Last Sequence Received Last Sequence Applied Difference
---------- ---------------------- --------------------- ----------
         1                      9                     7          2
```

STEP 2: Start MRP (Managed Recovery Process) on Standby DB

To start log apply process in real-time recovery mode:

```
SQL> ALTER DATABASE RECOVER MANAGED STANDBY DATABASE DISCONNECT FROM SESSION;
Database altered.
```

STEP 3: Verify Log Synchronization

Re-run gap query to verify sequence difference becomes 0:

```
SQL> SELECT ARCH.THREAD# "Thread",
       ARCH.SEQUENCE# "Last Sequence Received",
       APPL.SEQUENCE# "Last Sequence Applied",
       (ARCH.SEQUENCE# - APPL.SEQUENCE#) "Difference"
FROM
  (SELECT THREAD#, SEQUENCE# FROM V$ARCHIVED_LOG WHERE (THREAD#, FIRST_TIME) IN
    (SELECT THREAD#, MAX(FIRST_TIME) FROM V$ARCHIVED_LOG GROUP BY THREAD#)) ARCH,
  (SELECT THREAD#, SEQUENCE# FROM V$LOG_HISTORY WHERE (THREAD#, FIRST_TIME) IN
    (SELECT THREAD#, MAX(FIRST_TIME) FROM V$LOG_HISTORY GROUP BY THREAD#)) APPL
WHERE ARCH.THREAD# = APPL.THREAD# ORDER BY 1;

    Thread Last Sequence Received Last Sequence Applied Difference
---------- ---------------------- --------------------- ----------
         1                      9                     9          0
```

SUCCESS: The Primary Database (DG11) and Physical Standby Database (DG11S) are now fully synchronized in Real-Time Apply mode!